A Synopsis

Of

Project

On

TrafficSense: LLM-Based Urban Incident Intelligence

Submitted in partial fulfilment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE & ENGINEERING

Under the Guidance of

Narayan Chaturvedi

Professor

Project Team ID: MP2026CSE126

Submitted By

Name of Student: Manvi Jain	Enrollment No.: 232024229
Name of Student: Ayushman Raj	Enrollment No.: 232023764
Name of Student: Anshika Agarwal	Enrollment No.: 232023254
Name of Student: Nayan Kumar	Enrollment No.: 232023779

Academic Session: 2026–27

CERTIFICATE

This is to certify that the project synopsis entitled “TrafficSense: LLM-Based Urban Incident Intelligence” is being submitted by Manvi Jain (232024229), Ayushman Raj (232023764), Anshika Agarwal (232023254) and Nayan Kumar (232023779) for the partial fulfillment of the requirement for the award of the degree of B.Tech-CSE in Department of Computer Science & Engineering, Graphic Era Deemed to be University, Dehradun, and is a record of their own work carried out under my supervision. The matter embodied in the synopsis has not been submitted in any other university or institution for the award of any degree.

Narayan Chaturvedi
Professor

Department of Computer Science and Engineering,
Graphic Era Deemed to be University, Dehradun, India.

DECLARATION

We hereby declare that the project synopsis entitled “TrafficSense: LLM-Based Urban Incident Intelligence” is being submitted for the partial fulfillment of the requirement for the award of the degree of B.Tech-CSE in Department of Computer Science & Engineering, Graphic Era Deemed to be University, Dehradun, is our original work and has not been submitted to any other university or institution for the award of any degree or diploma.

Manvi Jain (232024229)	Signature: MANVI_JAIN
Ayushman Raj (232023764)	Signature: AYUSHMAN_RAJ
Anshika Agarwal (232023254)	Signature: ANSHIKA_AGARWAL
Nayan Kumar (232023779)	Signature: NAYAN_KUMAR

TABLE OF CONTENTS

S.No.	CONTENT	PAGE NO.
	Abstract	5
1.	Introduction	6
2.	Literature Review	7
3.	Problem Statement & Objectives	8
4.	Methodology Flowchart	9
5.	Proposed Methodology	10
6.	Hardware and Software Requirements	12
7.	Expected Outcomes	13
8.	Conclusion	14
	References	15

ABSTRACT

Urban mobility networks in developing nations experience recurring disruptions caused by non-recurrent events such as multi-vehicle collisions, monsoon waterlogging, sudden slope failures (landslides), civil demonstrations, and uncoordinated public works. Traditional Intelligent Transportation Systems (ITS) rely on capital-intensive hardware infrastructure—primarily inductive pavement loop detectors and CCTV cameras—that cost between $15,000 and $50,000 per intersection, cover at best 8% to 12% of metropolitan arterials, demand continuous civil maintenance, and output scalar metrics devoid of root-cause context.

This project proposes TrafficSense, an autonomous urban incident intelligence and predictive mobility analytics system that treats the public web, open civic data, and free-tier spatial telemetry as a distributed sensor network. The system continuously ingests unstructured text feeds, filters conversational noise, parses structured incident attributes via Large Language Models (LLMs), clusters reports into canonical events using PostGIS, and models physical clearance durations to deliver reliable route intelligence without capital expenditure.

The proposed architecture introduces a decoupled Two-Mode operational design: a lightweight National Macro-Sentry that records major nationwide disruptions into a persistent spatial archive, and an on-demand Hyperlocal Scout that evaluates en-route incidents by contrasting commuter travel time against dynamic queuing clearance rates. The resulting intelligence is presented through an interactive geospatial map dashboard and a safe, template-based conversational Text-to-SQL query assistant. The expected outcome is a zero-hardware, low-cost incident detection and route-advisory platform capable of national-scale situational awareness and hyperlocal decision support.

1.
INTRODUCTION

Standard ITS frameworks model vehicular flows using deterministic fluid analogies, such as Lighthill-Whitham-Richards hydrodynamic flow theory, or adaptive traffic signal algorithms such as Max-Pressure or Model Predictive Control. While mathematically rigorous under stationary conditions, these approaches rely on fixed roadway parameters—most notably static link saturation flow and lane capacity—that break down when non-recurrent events occur. An overturned commercial freight vehicle or localized flash flooding reduces effective link capacity by an arbitrary percentage, and because traditional sensor networks cannot report why a corridor has slowed, adaptive controllers and navigation routing algorithms routinely allocate green-time or route vehicular volumes directly into impassable bottlenecks.

Meanwhile, commercial social media APIs operate behind expensive paywalls, preventing civic agencies and academic researchers from utilizing public incident reports at scale. TrafficSense addresses this gap by combining free-tier data sources, open spatial telemetry, and LLM-based natural language understanding into a single incident-detection and route-advisory pipeline.

The proposed project adapts a hierarchical separation of concerns in which LLM reasoning is used strictly to extract physical parameters from unstructured news and citizen reports, rather than to directly control safety-critical infrastructure. These extracted parameters populate deterministic queuing equations, empirical duration waterfalls, and dynamic route-cost engines, ensuring that inference latency and hallucination risk never enter real-time routing or signal-control decisions. The overall architecture therefore moves from raw multi-source text and telemetry ingestion toward Ingest → Filter → Extract → Cluster → Model Clearance → Present Route Intelligence, providing a practical and measurable framework for low-cost, software-only traffic incident intelligence.

2.
LITERATURE REVIEW

Recent foundational research by Wei et al. (National University of Singapore, 2026), titled “Virtual Traffic Police: Large Language Model-Augmented Traffic Signal Control for Unforeseen Incidents,” establishes the mathematical bridge connecting unstructured linguistic incident descriptions to physical transportation dynamics. The study evaluates the operational trade-offs of deploying LLMs in safety-critical transportation domains and identifies three structural flaws in naive applications: inference latency bottlenecks, since direct closed-loop traffic control demands sub-second phase allocations while state-of-the-art LLM token generation introduces 1.5 to 4.0 seconds of latency; hallucination in critical decisions, since generative foundation models lack strict physical grounding and can hallucinate conflicting control parameters; and prohibitive infrastructure replacement costs associated with edge AI computing units.

To resolve these flaws, Wei et al. validate a two-level hierarchical framework in which an Upper Level LLM agent parses natural-language incident narratives, retrieves historical operational exemplars via a specialized Traffic Language Retrieval System (TLRS), and dynamically tunes physical parameters such as saturation flow rates or pressure weights, while lower-level deterministic controllers execute high-frequency signal switches using this updated parameter matrix. In microscopic SUMO simulation experiments under saturated morning-peak conditions, this hybrid approach reduced average vehicle delay by over 23% and achieved zero-second intersection delay for emergency vehicle preemption.

TrafficSense adapts this hierarchical separation of concerns to a non-safety-critical, advisory context. Rather than attempting direct hardware control of municipal traffic signals, the proposed system uses LLM reasoning purely to extract structured physical parameters—incident type, severity, blocked lanes, direction, and location—from unstructured news, civic data, and citizen reports. This separation ensures that LLM inference latency and hallucination risk are confined to an offline extraction stage and never propagate into real-time route or signal decisions, which instead rely on deterministic queuing mathematics and spatial clustering.

The research gap addressed by this project is the integration of free, open, multi-source ingestion; LLM-based structured extraction; native spatial deduplication via PostGIS; heterogeneous mixed-traffic (PCU) clearance modeling; and a safe, parameterized Text-to-SQL conversational interface into a single experimentally deployable incident-intelligence platform for Indian road conditions, where informal landmark references, extreme vehicle heterogeneity, and monsoon- and terrain-driven hazards are not well served by conventional Western ITS literature or commercial APIs.

3.
PROBLEM STATEMENT & OBJECTIVES

3.1 Problem Statement

Existing Intelligent Transportation Systems depend on capital-intensive hardware such as inductive pavement loop detectors and CCTV cameras, which are expensive to deploy, cover only a small fraction of metropolitan arterials, require continuous maintenance, and report only scalar occupancy or count metrics with no root-cause context. Commercial social media APIs that could otherwise supply citizen incident reports operate behind expensive paywalls, restricting access for civic agencies and academic researchers. As a result, non-recurrent disruptions—collisions, landslides, waterlogging, protests, and road closures—are detected late or not at all, and navigation systems continue routing commuters into impassable bottlenecks.

There is therefore a need for a zero-hardware, low-cost incident intelligence system that can continuously monitor open, free-tier text and spatial data sources; distinguish genuine road disruptions from conversational noise; extract structured incident attributes; deduplicate multiple reports of the same event; model realistic queue clearance durations for heterogeneous Indian traffic; and deliver this intelligence to commuters and civic operators through an interactive map and a safe conversational query interface.

3.2 Objectives

1. To collect traffic-related incident data from different sources, including news, civic feeds, police advisories, and other open data sources.
2. To use a fine-tuned Large Language Model (LLM) to remove noisy or irrelevant information, identify genuine traffic incidents, and answer traffic-related user queries.
3. To implement geocoding and spatiotemporal clustering to map incident reports to precise locations and combine multiple reports of the same incident.
4. To develop an analytics engine that studies mixed traffic using Passenger Car Units (PCU) and estimates how long an active disruption may take to clear using a three-level method.
5. To design an interactive geospatial dashboard with real-time heatmaps and route suggestions for commuters and traffic command centers.

4.
METHODOLOGY FLOWCHART

The proposed methodology is organized into five stages, moving from the overall system architecture through geocoding, LLM-based filtering and user interaction, analytics, and finally presentation. The flow below summarizes how these five stages connect end to end:

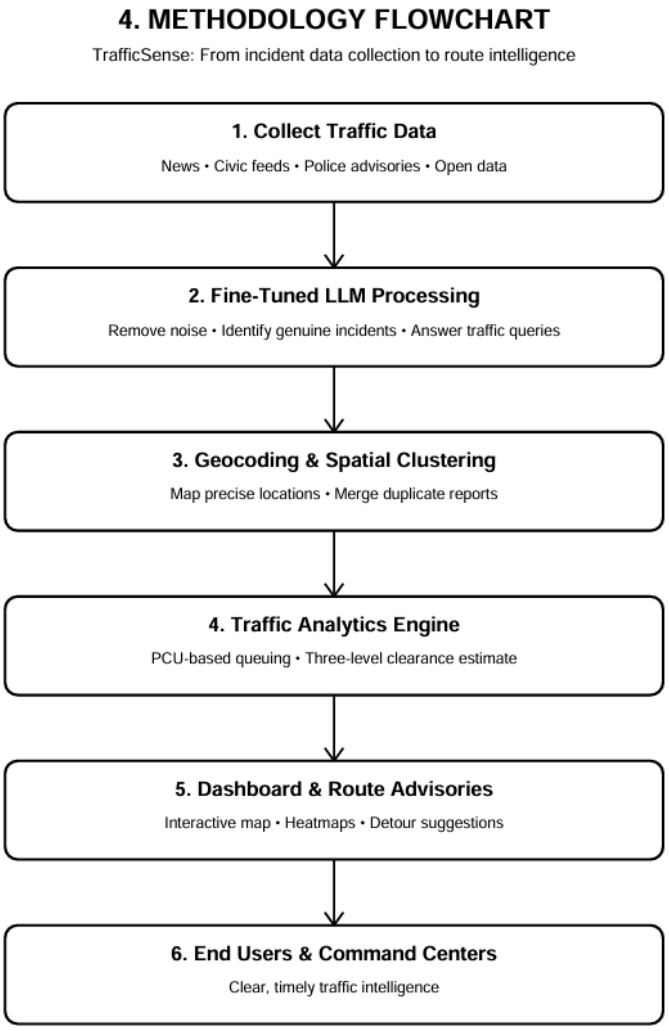


Figure 4.1: End-to-End Five-Stage Methodology Flowchart for TrafficSense

5.
PROPOSED METHODOLOGY

5.1 Overview of the Approach

The proposed system consists of the following major components, orchestrated through an internal Redis Streams event bus that decouples ingestion I/O, natural language inference, spatial clustering, and visualization into asynchronous worker threads:

- Multi-Source Ingestion Module (Google News RSS, GDELT, civic 311 data, police advisory mirrors, OpenStreetMap, Open-Meteo, TomTom/HERE)
- Relevance and Temporal Classification Module
- LLM-Based Structured Extraction Module (Pydantic-enforced schema)
- Cache-First Geocoding Module (local GeoNames gazetteer with fuzzy matching, rate-limited Nominatim fallback)
- PostGIS Spatiotemporal Clustering and Golden Record Synthesis Module
- Mixed-Traffic (PCU) Analytics and Three-Level Waterfall Clearance Engine
- Hyperlocal Choke-Point Detection and En-Route Temporal Overlap Assessment Module
- Presentation Layer: React-Leaflet Map Canvas and Parameterized Text-to-SQL Conversational Assistant

Raw items are pushed onto a raw-items stream by ingestion workers, passed through relevance and temporal classification, structured by the LLM extractor into a validated JSON schema, and finally geocoded, clustered, and committed as canonical events in PostGIS. Unrecoverable failures at any stage are redirected to a dead-letter stream, ensuring no article is silently lost.

5.2 Two-Mode Operational Design

The system operates in two decoupled modes. The National Macro-Sentry is a lightweight background engine that polls free-tier sources roughly once every two hours, applies a catastrophic-severity gatekeeper to discard minor conversational complaints, and commits only major verified disruptions (fatal collisions, landslides, bridge washouts, highway blockades, or severe waterlogging) into a persistent spatial archive. A startup Catch-Up Engine detects downtime gaps using a stored heartbeat timestamp and backfills missed events from historical news endpoints before real-time polling resumes.

The Hyperlocal Scout is an on-demand engine that activates only when a commuter or operator queries a specific route. It computes the route geometry, buffers it spatially in PostGIS, identifies the top three to five topological choke points along the corridor using barrier tags, lane reductions, terrain slope, and historical hotspots, and samples live speed-flow telemetry only at those points—conserving free-tier API quotas while still enabling temporal overlap assessment between commuter arrival time and predicted incident clearance time.

5.3 Structured Information Extraction

Incoming text passes through a two-stage classification strategy: a relevance classifier tuned for high recall to separate metaphorical expressions from genuine roadway disruptions, and a temporal disambiguator that determines whether an incident is still active or already cleared. To balance development speed with academic rigor, the pipeline begins with zero-shot prompting on a lightweight foundation model and progressively distills a local SetFit (Sentence Transformer Few-Shot) classifier from logged predictions and human-validated examples, enabling a formal comparison between zero-shot and distilled classification accuracy. Geo-parsing is handled in two stages: an LLM-based linguistic normalizer converts colloquial landmark references into a clean geographic tuple, followed by a structured schema enforcer that extracts standardized attributes such as incident type, severity, lanes blocked, direction, and timestamp.

5.4 Deduplication and Clearance Modeling

Multiple reports of the same incident are merged using PostGIS native spatial clustering (ST_ClusterDBSCAN) with a 500-meter spatial threshold and a 60-minute rolling temporal window, and a single Golden Record is synthesized per cluster based on attribute completeness. Queue loads and discharge capacities are standardized using Passenger Car Units (PCU) to account for India's heterogeneous vehicle mix, and clearance duration is estimated through a three-level fallback waterfall: empirical corridor history, spatial-neighbor imputation from similar road segments, and, in the absence of any prior data, deterministic queuing or volumetric debris-clearance mathematics. Predictions are dynamically bounded using a Bayesian update rule with type-specific hard ceilings to avoid unrealistic or stale duration estimates.

5.5 Presentation and Query Interface

The system presents incident intelligence through a React-Leaflet map dashboard with color-coded severity markers and a historical timeline slider, alongside a conversational assistant built on an Intent Classification and Slot-Filling architecture rather than free-form SQL generation. This design maps user queries to a fixed set of parameterized PostGIS query templates, avoiding SQL injection risk and geometry hallucination while still supporting natural-language questions about nearby incidents, corridor history, average clearance times, and congestion hotspots.

6.
HARDWARE AND SOFTWARE REQUIREMENTS

6.1 Hardware Requirements

• A laptop or workstation with a reasonably capable CPU.

• Intel Core i5/i7 or equivalent processor.

• Minimum 16 GB RAM recommended.

• GPU is optional but recommended for local SetFit classifier training and inference.

• High-speed internet connection for free-tier API access (news, weather, geocoding, telemetry).

• Adequate local storage for the PostGIS database, pre-seeded gazetteer (~300,000 place names), and logs.

6.2 Software Requirements

• Python 3.11+

• FastAPI

• PostgreSQL 16+ with PostGIS extension

• Redis (Streams)

• React and Leaflet (react-leaflet)

• APScheduler with asyncio

• Pydantic for schema enforcement

• rapidfuzz for fuzzy gazetteer matching

• SetFit / Sentence Transformers

• OSRM and OpenStreetMap Overpass API

• Git and GitHub

7.
EXPECTED OUTCOMES

By the end of this project, the following outcomes are expected:

1. A complete zero-hardware urban incident intelligence and predictive mobility analytics system.
2. An automated multi-source ingestion and LLM-based filtering pipeline converting unstructured public feeds into validated incident records.
3. A functioning PostGIS-based spatiotemporal deduplication and Golden Record synthesis engine for canonical incident mapping.
4. A heterogeneous mixed-traffic (PCU) clearance-duration model with a three-level fallback waterfall and dynamic bounds.
5. An interactive geospatial map dashboard with real-time heatmaps, route advisories, and a safe parameterized conversational query assistant.

Evaluation Measures

• Relevance classification recall and precision (zero-shot vs. distilled SetFit).

• Structured extraction accuracy against manually annotated incident records.

• Spatial clustering correctness and Golden Record completeness.

• Clearance-duration prediction error across the three waterfall levels.

• API and database resource consumption relative to free-tier quotas.

• End-to-end pipeline latency from ingestion to map presentation.

8.
CONCLUSION

The proposed TrafficSense framework addresses a critical limitation of conventional Intelligent Transportation Systems: their dependence on expensive, sparsely deployed hardware sensors that report only scalar traffic metrics with no context on why a corridor has slowed. By treating the open web, civic data, and free-tier spatial telemetry as a distributed sensor network, TrafficSense enables national-scale situational awareness and hyperlocal route intelligence without capital expenditure on physical infrastructure.

The proposed architecture combines multi-source ingestion, two-stage relevance and temporal classification, LLM-based structured extraction, native spatial deduplication, heterogeneous mixed-traffic clearance modeling, and a safe parameterized conversational interface into a single, decoupled, event-driven pipeline. By separating LLM reasoning (used for offline parameter extraction) from deterministic real-time routing and queuing mathematics, the system avoids the inference-latency and hallucination risks associated with direct LLM control of safety-critical infrastructure.

The Two-Mode operational design—a background National Macro-Sentry and an on-demand Hyperlocal Scout—further ensures that free-tier API and database quotas are respected while still delivering timely, actionable route advisories. The resulting implementation is expected to serve as a practical, measurable, zero-cost alternative to hardware-dependent ITS deployments, with clear extensions toward edge vision-language verification, spatio-temporal graph neural networks, and dynamic in-vehicle navigation.

REFERENCES

[1] Wei et al., “Virtual Traffic Police: Large Language Model-Augmented Traffic Signal Control for Unforeseen Incidents,” National University of Singapore, 2026.

[2] Lighthill, M. J. and Whitham, G. B., “On Kinematic Waves II: A Theory of Traffic Flow on Long Crowded Roads,” Proceedings of the Royal Society A, 1955.

[3] Varaiya, P., “Max Pressure Control of a Network of Signalized Intersections,” Transportation Research Part C, 2013.

[4] Indian Road Congress (IRC), “Guidelines for Capacity of Urban Roads in Plain Areas,” IRC:106.

[5] Ester, M. et al., “A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise (DBSCAN),” KDD, 1996.